FEMSSISA Grade 6 Olympiad

Complete 6-Month Training Programme

You finished Grade 5. Now it gets even more exciting.

This programme takes you from where you are right now all the way to competition-ready in 6 months.

How to use it: Read 1 step at a time. Don't rush. Do the practice questions. Come back the next day and read the previous step for 5 minutes before starting the new one.

Papers analysed: 2022, 2023, 2024, 2025 (plus 2017 Final Round as a bonus challenge).

Official past papers and answers: www.saolympiads.co.za | **Certainty: 9/10**

🌟 **BEFORE YOU BEGIN — READ THIS ONCE**

The Grade 6 paper has **20 questions** in 90 minutes. Questions 1–15 are multiple choice (1 mark each). Questions 16–20 are open-ended (2 marks each). Total: 25 marks.

The three golden rules under exam conditions:

1. Read every question **twice** before writing anything.
2. Show all your working — even if you think the answer is obvious.
3. Check by plugging your answer back into the original question.

What Is New in Grade 6 vs Grade 5?

Grade 6 keeps all the Grade 5 topics but adds new ones on top. The table below shows you what is brand new, and why each one is slightly trickier.

New topic in Grade 6	Why it needs a new skill
Decimal division to specified decimal places	Needs long division and a careful rounding step
Algebraic factoring (distributive law)	Spot a shared number, factor it out, simplify fast
Divisibility rules for 3, 6, 9, 12	Find a missing digit using digit-sum rules
Cryptarithmetic (letters = digits in addition sums)	Column-by-column logic with carrying
Composite area (L-shapes and step shapes)	Split or subtract rectangles to find areas
Grid polygon area (diagonal shapes on a grid)	Surround with a rectangle, subtract the corners
Lines of symmetry with placed squares	Visual folding logic
Cube tower patterns	Triangular numbers and their running totals
Hotel room positioning	Move along a number line and work backwards
Strike-off sequences	Two rounds of crossing off, then count survivors
Watch that loses or gains time	Proportional reasoning between two time-streams
Plant geometric growth (multiply each week)	Go backwards through repeated multiplication
Sum of cubes formula	Square of a triangular number — beautiful shortcut
Telescoping series (lots of fractions that cancel)	Spot the identity, split each term, watch it collapse
Worst-case probability (blindfolded discs)	The word "ensure" means assume the worst luck always
Three-variable equations (basketball)	Three equations, substitute step by step

Question Type Frequency Map (2022–2025)

This is your treasure map. Topics that appear **EVERY YEAR** should get the most practice time.

Topic	2022	2023	2024	2025	Priority
BODMAS with fractions	Q1	Q1	Q1	Q1	⭐ **EVERY YEAR**
Decimal division & rounding	Q2	Q2	Q2	Q2	⭐ **EVERY YEAR**
Algebraic simplification & factoring	Q3	Q3	Q3,4,6	Q3,4,5,6	⭐ **EVERY YEAR**
Container and capacity fractions	Q5	Q5	Q8	Q8	⭐ **EVERY YEAR**
Rate, proportion and price	Q6	Q6	Q7	Q7	⭐ **EVERY YEAR**
Divisibility and missing digit	Q7	Q7	Q9	Q9	⭐ **EVERY YEAR**
Age problems	Q12	Q12	Q14	Q14	⭐ **EVERY YEAR**
Grid polygon area	Q13	Q13	Q15	Q15	⭐ **EVERY YEAR**
Hotel room positioning	Q15	Q15	Q17	Q17	⭐ **EVERY YEAR**
Cube tower patterns	—	Q10	Q12	Q12	⭐ **EVERY YEAR**
Cryptarithmetic	Q4	Q4	—	—	★ High
Midpoint fractions	Q8	Q19	—	—	★ High
Profit and buy-sell coins	Q10	—	—	Q10	★ High
Watch loses or gains time	Q17	—	Q13	Q13	★ High
Worst-case disc probability	—	Q8	Q10	—	★ High
Related rectangles	—	Q9	Q11	Q11	★ High
Fraction equation: find n	—	Q14	Q16	Q16	★ High
Strike-off sequences	—	Q16	Q18	Q18	★ High
Plant geometric growth	—	—	Q19	Q19	★ High
Sum of cubes	—	Q20	Q20	—	★ High
Telescoping fractions	Q20	—	—	—	Medium
Lines dividing a plane	—	—	—	Q20	Medium

6-Month Overview Schedule

Month	Phase	What you learn	Steps
Month 1	Foundation Repair	BODMAS, decimals, fractions, basic algebra	1–4
Month 2	Core Number	Divisibility, cryptarithmetic, rates and profit	5–7
Month 3	Geometry	Composite area, grid area, symmetry, cube towers	8–11
Month 4	Logic & Problem Solving	Age, hotel rooms, strike-offs, watch time, plant growth	12–16
Month 5	Advanced Topics	Series, probability, three-variable equations	17–20
Month 6	Paper Practice	Full timed papers, 2017 bonus, final revision	—

🧠 **YOUR WEEKLY ROUTINE**

Session 1 (45 min): Read the new step slowly. Talk through each example out loud as if you are teaching someone else. This is the single best way to check you actually understand it.

Session 2 (45 min): Close the booklet and try to redo the worked examples from memory. Then do the practice questions at the end of the step.

Session 3 (45 min): Mixed review — redo one question from each previous step, plus use the online resource listed for that week.

🔑 FOUNDATION CONCEPT A — VARIABLES AND THE BALANCE METHOD

Read this before any step. Algebra is the tool behind almost every problem in this programme. Once you own this idea, everything else unlocks.

What is a Variable?

📖 **WHAT IT IS**

A **variable** is a letter that holds the place of a number you have not found yet. It is like a box with a question mark drawn on it. You write a letter — usually *n*, *x*, or *A* — and that letter means "the number I am looking for."

Example: *"I think of a number. I multiply it by 3, then add 5. I get 20."*

In letter form: **3n + 5 = 20**. The letter n is our mystery box.

What is an Equation?

📖 **WHAT IT IS**

An equation is a statement that two things are exactly equal — like a perfectly balanced scale. The equals sign (=) is the pivot in the middle. Whatever is on the left weighs exactly the same as whatever is on the right.

The golden rule: Whatever you do to one side of the scale, you MUST do the exact same thing to the other side. Otherwise the scale tips and becomes unbalanced (wrong).

🧠 **PICTURE THIS**

Imagine an old-fashioned balance scale in front of you. The left pan has "3n + 5" on it. The right pan has "20" on it. Right now they balance perfectly. Every step you take must keep the scale balanced. If you subtract 5 from the left pan, you must subtract 5 from the right pan too.

Solving — the Balance Method, Step by Step

📝 **WORKED EXAMPLE: SOLVE 3N + 5 = 20**

Goal: get n alone on one side.

Step 1 — Remove the +5. The opposite of +5 is −5. Subtract 5 from BOTH sides:

```
3n + 5 − 5 = 20 − 5
3n = 15
```

Step 2 — Remove the ×3. The opposite of ×3 is ÷3. Divide BOTH sides by 3:

```
3n ÷ 3 = 15 ÷ 3
n = 5
```

Step 3 — Check: substitute n = 5 back in the original:

```
3(5) + 5 = 15 + 5 = 20 ✓
```

⚡ **THE TWO UNDO RULES**

- To undo **adding** something → **subtract** it from both sides.
- To undo **subtracting** something → **add** it to both sides.
- To undo **multiplying** by something → **divide** both sides by it.
- To undo **dividing** by something → **multiply** both sides by it.

Mnemonic — "UNDO the last step first." Work backwards through what was done to n, undoing each operation one at a time.

✏️ **QUICK PRACTICE — SOLVE THESE**

1. 4n + 3 = 19
2. 2n − 7 = 11
3. n ÷ 3 + 1 = 6
4. "I think of a number, multiply by 5 and subtract 8. I get 42. What is the number?"

(Answers: n=4, n=9, n=15, n=10)

🔑 FOUNDATION CONCEPT B — ADDING UP A SEQUENCE (THE GAUSS TRICK)

This single idea appears in at least 4 different question types in Grade 6. Learn it once and it pays off everywhere.

What is an Arithmetic Sequence?

📖 **WHAT IT IS**

An **arithmetic sequence** is a list of numbers where each new number is made by adding the same fixed amount to the previous one. That fixed amount is called the **common difference**.

- 1, 2, 3, 4, 5, 6 … (common difference = 1) — the counting numbers
- 2, 4, 6, 8, 10 … (common difference = 2) — the even numbers

- 11, 13, 15, 17 ... (common difference = 2, starts at 11)
- 5, 10, 15, 20 ... (common difference = 5)

An **arithmetic series** is the sum of all the terms in such a list.

The Gauss Story — and Why It Works

📖 THE STORY

About 200 years ago, a young boy named Carl Friedrich Gauss was given a punishment by his school teacher: add all the numbers from 1 to 100. The teacher expected this to take the whole lesson. Gauss finished in about 10 seconds. Here is how he did it.

📝 DERIVING THE FORMULA: 1 + 2 + 3 + … + N

Write the sum forwards. Below it, write the same sum backwards. Now add them vertically:

```
Forward: 1 + 2 + 3 + … + (n−1) + n
Backwards: n + (n−1) + (n−2) + … + 2 + 1
──────────────────────────────────────────
Each pair: (n+1) + (n+1) + (n+1) + … + (n+1)
```

There are n pairs, and each pair adds to (n+1).

```
Two rows total = n × (n+1)
```

But we wrote the sum TWICE, so divide by 2:

```
Sum = n × (n+1) ÷ 2
```

Gauss's answer: 1+2+...+100 = 100 × 101 ÷ 2 = **5 050**. Ten seconds. Done.

```
Sum of 1 + 2 + 3 + … + n = n(n+1) ÷ 2
```

🧠 VERIFY WITH SMALL NUMBERS FIRST

```
n=4: 1+2+3+4 = 10. Formula: 4×5÷2 = 10. ✓
n=5: 1+2+3+4+5 = 15. Formula: 5×6÷2 = 15. ✓
n=10: 1+2+…+10 = 55. Formula: 10×11÷2 = 55. ✓
```

Always verify a formula with small numbers you can check by hand. If it works for n=3, n=4 and n=5, you can trust it.

Triangular Numbers — a Beautiful Pattern

📖 WHAT IT IS

The numbers 1, 3, 6, 10, 15, 21, 28 ... are called **triangular numbers** because you can arrange that many dots into a triangle shape (like the pins in bowling). The nth triangular number is exactly n(n+1)/2 — our formula! You will see these in cube tower problems and the sum of cubes formula.

The General Formula (When the Sum Doesn't Start at 1)

```
Sum = (number of terms) × (first term + last term) ÷ 2
```

📝 **EXAMPLE: ADD ALL ODD NUMBERS FROM 11 TO 69**

Step 1 — Count the terms: `11, 13, 15, …, 69. Number of terms = (69−11)÷2 + 1 = 30.`

Step 2 — Apply the formula:

`Sum = 30 × (11 + 69) ÷ 2 = 30 × 80 ÷ 2 = 30 × 40 = 1 200`

⚡ **QUICK TRICKS — WHEN TO USE THIS FORMULA**

- Any question asking you to add a long run of numbers → Gauss formula.
- Cube tower totals → sum of triangular numbers (Step 11).
- Sum of cubes → square of a Gauss sum (Step 17).
- Lines dividing a plane → Gauss sum + 1 (Bonus E).
- **Mnemonic: "Pair the ends, count the middles."** First + last, multiplied by how many pairs, divided by 2.

PHASE 1 — FOUNDATION REPAIR | MONTH 1, WEEKS 1–4

Questions 1–6 of every paper. The easiest marks to earn. The easiest to lose through rushing. Master these first and you are already ahead of most of the competition.

STEP 1 BODMAS with Fractions and Decimals

What it is

📖 **WHAT IT IS AND WHY IT MATTERS**

Every year Question 1 and Question 5 test whether you do operations in the right order. If you just work left-to-right without thinking, you get the wrong answer every time — even though the arithmetic is easy.

The rule is called **BODMAS**. Each letter tells you which operation to do first:

Letter	Stands for	Symbol
B	Brackets — do these first, always	( )
O	Orders — powers and roots	^{2} √
D	Division — same level as M	÷
M	Multiplication — do D and M left to right	×
A	Addition — same level as S	+
S	Subtraction — do A and S left to right	–

The most important rule on this entire exam: × and ÷ happen BEFORE + and –. This is the rule that trips up the most students.

🧠 **MNEMONIC — SAY THIS OUT LOUD RIGHT NOW**

"Big Orange Dogs Might Act Strange." (Brackets, Orders, Division, Multiplication, Addition, Subtraction.)

Write it on a sticky note. Say it five times. Now you will never forget the order.

D and M are twins: They are the same level of importance. When you see both in the same calculation, go *left to right* — whichever comes first in the expression, do that one first.

The Method — 3 Steps

⚡ THE 3-STEP SYSTEM

1. **Circle** any brackets. Solve whatever is inside them first.
2. **Underline** all × and ÷ signs. Solve those next, working left to right.
3. Now do all + and − from left to right. You're done.

Worked Examples

📝 2022 Q1: EVALUATE 16 − (1/4) × 16

Step 1 — Underline the ×: (1/4) × 16. Do this first.

```
(1/4) × 16 = 4
```

Step 2 — Now do the subtraction:

```
16 − 4 = 12
Answer: (C)
```

🚫 *Common mistake: doing 16 − 1/4 = 15.75 first, then ×16 = 252. Wrong!*

✅ *Always multiply BEFORE subtracting.*

📝 2025 Q5 — DECIMAL BODMAS: EVALUATE 8 ÷ 0.04 × 0.2

Both ÷ and × are the same level. Go left to right.

Step 1 — The ÷ is first (leftmost): 8 ÷ 0.04

Shortcut for ÷ by a decimal: multiply both by 100 to remove the decimal point:

```
8 ÷ 0.04 → multiply top and bottom by 100 → 800 ÷ 4 = 200
```

Step 2 — Then the ×:

```
200 × 0.2 = 40
Answer: (A)
```

🧠 THE DECIMAL DIVISION SHORTCUT

Dividing by 0.04 looks scary. Here's how to make it easy: **multiply both numbers by the same power of 10** to turn the divisor into a whole number.

- ÷ 0.04 → multiply both by 100 → ÷ 4 (easy!)
- ÷ 0.2 → multiply both by 10 → ÷ 2 (easy!)
- ÷ 0.005 → multiply both by 1000 → ÷ 5 (easy!)

Think of it like a fraction: 8/0.04 = 800/4. You scaled the top and bottom by the same amount, so the answer doesn't change — just like with fractions.

⚡ QUICK TRICKS — BODMAS

- See a × or ÷ anywhere? Do it before any + or − near it.
- See ÷ by a decimal? Multiply both numbers by 10 (or 100) first.
- When ÷ and × appear together, go strictly left to right.
- Q1 always tests this. Take 30 seconds and do it carefully. Do not rush.

✏️ PRACTICE QUESTIONS

1. 20 − 3 × 5
2. 4 + 6 × 2 − 1
3. 0.9 ÷ 0.03 × 0.1
4. 15 + (1/3) × 15
5. 12 ÷ 0.4 × 0.5

(Answers: 5, 15, 3, 20, 15)

STEP 2 Decimal Division and Rounding

What it is

📖 WHAT IT IS AND WHY IT MATTERS

Question 2 is in every single paper. You are asked to divide one number by another and round the answer to a specific number of decimal places. The method never changes. Once you know it, Q2 is always a free mark.

Key language to know:

- "Correct to 2 decimal places" → you want 2 digits after the decimal point.
- "Round off to 1 decimal place" → you want 1 digit after the decimal point.

The rounding rule: To round to 2 places, you actually need to calculate 3 places. Then look at the 3rd digit. If it is 5 or more — add 1 to the 2nd digit. If it is 4 or less — leave the 2nd digit exactly as it is.

🧠 MNEMONIC FOR ROUNDING

"5 and above, round it up. 4 and below, let it go."

Picture a round hill. Numbers 5, 6, 7, 8, 9 are past the top of the hill, so they roll upward. Numbers 0, 1, 2, 3, 4 are still on the way up, so they stay where they are.

The Method

⚡ THE 5-STEP SYSTEM

1. Write the dividend with three extra zeros: e.g. 13.000
2. Do long division, one digit at a time.
3. For each step: how many times does the divisor fit? Write that digit. Multiply. Subtract. Bring down the next zero.
4. Stop when you have *one more decimal place* than the question asks for.
5. Apply the rounding rule to the last digit you calculated.

📝 2022 Q2: CALCULATE 13 ÷ 7, CORRECT TO 2 DECIMAL PLACES

We need 3 decimal places so we can round to 2.

```
13.000 ÷ 7
  7 goes into 13 → once (7×1=7). Remainder = 13−7 = 6. Digit: 1
  Bring down zero: 60. 7 goes into 60 → 8 times (7×8=56). Remainder = 4. 1st decimal: 8
  Bring down zero: 40. 7 goes into 40 → 5 times (7×5=35). Remainder = 5. 2nd decimal: 5
  Bring down zero: 50. 7 goes into 50 → 7 times (7×7=49). Remainder = 1. 3rd decimal: 7
Result so far: 1.857
```

The 3rd decimal (7) is ≥ 5, so round UP the 2nd decimal (5 becomes 6):

```
Answer: 1.86 → (A)
```

📝 2025 Q2: CALCULATE 15 ÷ 8, CORRECT TO 1 DECIMAL PLACE

We need 2 decimal places so we can round to 1.

```
15.00 ÷ 8
  8 goes into 15 → once (8×1=8). Remainder = 7. Digit: 1
  Bring down zero: 70. 8 goes into 70 → 8 times (8×8=64). Remainder = 6. 1st decimal: 8
  Bring down zero: 60. 8 goes into 60 → 7 times (8×7=56). Remainder = 4. 2nd decimal: 7
Result so far: 1.87
```

Rounding to 1 decimal place: look at 2nd decimal (7). 7 ≥ 5, so round UP the 8 to 9:

```
Answer: 1.9 → (B)
```

🧠 QUICK CHECK TRICK

After getting your answer, multiply back: 1.86 × 7 should give roughly 13. Let's check: 1.86 × 7 = 13.02. The tiny difference (0.02) comes from the remainder in the division — that is perfectly normal and confirms your answer is right.

✏️ **PRACTICE QUESTIONS**

1. 23 ÷ 7, correct to 2 decimal places
2. 17 ÷ 6, correct to 2 decimal places
3. 9 ÷ 4, correct to 1 decimal place
4. 22 ÷ 9, correct to 2 decimal places

(Answers: 3.29, 2.83, 2.3 (rounds to 2.3), 2.44)

STEP 3 Fractions Mastery — Four Types

Grade 6 uses fractions in four completely different ways. Study each type by itself. Each one has its own method.

Type A — Container and Capacity Fractions

📖 **WHAT IT IS AND WHY IT WORKS**

A container starts at one fraction full. Some liquid is removed (or added). It ends at a different fraction full. The amount of liquid that moved represents exactly the **difference** between those two fractions of the total capacity.

If a full tank holds C litres, and you go from 3/4 full to 2/3 full, you removed (3/4 − 2/3) × C litres. We know the amount, so we solve for C.

The key takeaway: the fraction change × capacity = amount moved. So capacity = amount ÷ fraction change.

```
Capacity = Amount moved ÷ (bigger fraction - smaller fraction)
```

🧠 **MNEMONIC — "THE FRACTION GAP"**

Imagine two water level marks on the side of the tank. One mark says "3/4" and the other says "2/3." The **gap between the marks** represents the fraction of capacity that was removed. You know the actual litres removed, so: litres ÷ gap fraction = full capacity.

Chant: "Amount over gap fraction. Amount over gap fraction."

📝 **2023 Q5: CONTAINER IS 3/4 FULL. REMOVE 120 LITRES. NOW 2/3 FULL.**

Step 1 — Find the fraction gap: `3/4 - 2/3 = 9/12 - 8/12 = 1/12`

Step 2 — 120 litres = 1/12 of the full capacity.

Step 3 — Solve: `Capacity = 120 ÷ (1/12) = 120 × 12 = 1 440 litres → (B)`

📝 2025 Q8: TANK IS 3/4 FULL. ADD 300 LITRES. NOW 5/6 FULL. (WATER ADDED, NOT REMOVED!)

The formula works exactly the same whether you add or remove.

Step 1 — Fraction gap: `5/6 − 3/4 = 10/12 − 9/12 = 1/12`

Step 2 — 300 litres = 1/12 of full capacity.

Step 3 — Capacity = 300 × 12 = ***3 600 litres*** → `(D)`

⚡ QUICK TRICKS — FRACTIONS TYPE A

- Always find a common denominator before subtracting the fractions.
- "Dividing by a fraction" = multiplying by its flip. So ÷ (1/12) = × 12.
- If the answer seems way too big — you probably forgot to flip the fraction. Check again.
- **Double-check:** multiply your capacity answer by each fraction. The difference should equal the amount removed.

Type B — Fraction Equations: Solve for n

📖 WHAT IT IS

You get an equation like (1/6) + (3/n) = 13/24 and must find the value of n. This is a balance scale problem — isolate the term with n, then solve for n directly.

📝 2023 Q14: (1/6) + (3/N) = 13/24. FIND N.

Step 1 — Move 1/6 to the right (subtract from both sides):

`3/n = 13/24 − 1/6`

Step 2 — Common denominator 24:

`3/n = 13/24 − 4/24 = 9/24 = 3/8`

Step 3 — Both sides have 3 as numerator, so denominators must match:

`n = 8` → `(D)`

📝 2025 Q16: (5/12) – (2/N) = 1/4. FIND N.

Step 1 — Isolate 2/n (add 2/n to both sides, subtract 1/4 from both sides):

`2/n = 5/12 − 1/4`

Step 2 — Common denominator 12:

`2/n = 5/12 − 3/12 = 2/12 = 1/6`

Step 3 — 2/n = 1/6. Cross-multiply:

`1 × n = 2 × 6 = 12`

🧠 **CROSS-MULTIPLICATION SHORTCUT**

When you have (something)/n = a/b, use cross-multiplication: multiply diagonally. The product of the two "cross" values are equal.

So if 2/n = 1/6 then: 2 × 6 = 1 × n → 12 = n. Quick!

Type C — Midpoint Fractions

📖 **WHAT IT IS**

A number is exactly halfway between two others. That means it is the average of the two end values. Average = (first + second) ÷ 2. Working backwards to find one end: End = 2 × Middle − Other end.

```
x = 2 × Middle - Other end
```

📝 **2022 Q8: 2/3 IS MIDWAY BETWEEN X AND 7/8. FIND X.**

```
x = 2 × (2/3) - 7/8 = 4/3 - 7/8
Common denominator 24: = 32/24 - 21/24 = 11/24 → (D)
```

Type D — Finding the Whole from a Part

📖 **WHAT IT IS**

You know a certain amount is a certain fraction of the total. Divide the amount by that fraction to get the total.

📝 **2022 Q5: VENDOR SOLD 1/4 OF POTATOES, THEN 35 MORE. NOW HAS 1/3 LEFT. TOTAL?**

After selling 1/4, the vendor has 3/4 left.

After selling 35 more, the vendor has 1/3 left.

The 35 represents the gap: 3/4 − 1/3 = 9/12 − 4/12 = 5/12 of the total.

```
Total = 35 ÷ (5/12) = 35 × 12/5 = 7 × 12 = 84 pockets → (C)
```

✏️ **PRACTICE QUESTIONS**

1. Tank is 2/3 full. Remove 600 litres. Now 1/4 full. What is the capacity?
2. Solve: (1/4) + (2/n) = 5/12. Find n.
3. 3/5 is midway between x and 9/10. Find x.
4. A vendor sold 1/3 of apples, then 40 more. Now has 1/4 left. How many apples?

(Answers: 1440L, n=12, x=3/10, 144 apples)

STEP 4 Algebraic Simplification — The Factoring Tricks

What it is

📖 **WHAT IT IS AND WHY IT WORKS**

Questions 3 and 4 in Grade 6 always give you a large multiplication expression. The trap is trying to calculate it directly — you will run out of time and make errors. The trick is **factoring out a common number**, which turns a scary calculation into a tiny one.

This works because of the **Distributive Law**:

$$a \times b + a \times c = a \times (b + c)$$

Reading it right-to-left: if you see two terms that share the same multiplied number (a common factor), you can pull that number outside a bracket. Then the bracket often simplifies to something tiny like 1 or 1000.

The key takeaway: before you calculate anything, scan every term and ask "does the same number appear in all of them?"

🧠 **THE FACTORING TRICK IN PICTURES**

Think of 156 × 156 − 156 × 155. Imagine 156 groups of 156 cookies, minus 156 groups of 155 cookies. You have 156 groups, and each group changed by (156 − 155) = 1 cookie. So you have 156 × 1 = 156 cookies total. The big numbers vanish!

The Method

⚡ **THE 4-STEP SYSTEM**

1. **Scan** every term in the expression. Look for the same number appearing everywhere.
2. **Factor out** that common number: common × (remaining terms).
3. **Simplify** inside the bracket — it is usually a tiny calculation.
4. **Multiply** the common factor by the bracket result.

📝 WORKED EXAMPLES FROM PAST PAPERS

2022 Q3: 156 × 156 – 156 × 155

```
Both terms share 156 as a factor.
= 156 × (156 − 155) = 156 × 1 = 156 → (B)
```

Without the trick: 24 336 – 24 180 = 156. Same answer, ten times more work!

2023 Q3: 201 × 174 – 172 × 201 – 201

```
Note: 201 = 201 × 1. All three terms share 201.
= 201 × (174 − 172 − 1) = 201 × 1 = 201 → (A)
```

2025 Q6: 201 × 58 – 29 × 2

Hidden common factor: 29 × 2 = 58. So rewrite:

```
= 201 × 58 − 58 = 58 × (201 − 1) = 58 × 200 = 11 600 → (C)
```

Trick 2 — Reaching a Round Number: 264 × 75

75 = 300 ÷ 4. So:

```
264 × 75 = 264 × 300 ÷ 4 = 79 200 ÷ 4 = 19 800 → (D)
```

The trick: replace 75 with (something ÷ 4) to reach 300, a round number.

⚡ QUICK TRICKS — FACTORING

- Before computing anything: scan every term for a shared number.
- 25 × anything → think: 25 = 100 ÷ 4. Multiply by 100 and divide by 4.
- 75 × anything → think: 75 = 300 ÷ 4. Multiply by 300 and divide by 4.
- If you spot "a × b – b", factor out b: b × (a – 1). The bracket is almost always 1 or a round number.
- **Mnemonic: "Same number? Factor it out!"**

✏️ PRACTICE QUESTIONS

1. 345 × 345 – 345 × 344
2. 201 × 99 – 99
3. 500 × 75 × 4
4. 88 × 25

(Answers: 345, 19 800, 150 000, 2 200)

STEP 5 Divisibility Rules and Missing Digit Problems

What it is

📖 **WHAT IT IS AND WHY IT MATTERS**

A number is "divisible by n" if n divides into it exactly with no remainder left over. Every divisor has a clever shortcut — instead of doing the full division, you just look at the digits of the number. This saves a lot of time.

In the exam, you will be given a number with one digit replaced by a letter (like m). You are told the remainder when the number is divided by 9 (or 12). You must find the value of m.

Divisor	The shortcut rule	Why it works
2	Last digit is even (0, 2, 4, 6 or 8)	Only the ones column determines evenness
3	Add all digits. If the sum divides by 3, so does the number	10 leaves remainder 1 when ÷3, so each digit contributes its face value
4	Last two digits form a number divisible by 4	100 is exactly ÷4, so higher digits don't matter
6	Passes BOTH the rule for 2 AND the rule for 3	6 = 2 × 3
9	Add all digits. If the sum divides by 9, so does the number	Same logic as 3, but for 9
12	Passes BOTH the rule for 3 AND the rule for 4	12 = 3 × 4

🧠 **MNEMONIC — "THE DIGIT SUM CLUB"**

The numbers **3** and **9** are members of the "Digit Sum Club." To test them, add up all the digits of the number. If your sum is in the club (divisible by 3 or 9), then the original number is too.

Example: Is 729 divisible by 9? 7+2+9 = 18. Is 18 divisible by 9? Yes (18÷9=2). So 729 is divisible by 9. ✓

The Method for Missing Digit Problems

⚡ **THE 4-STEP SYSTEM**

1. Write the digit sum (or last-digits expression) in terms of the missing digit m.
2. Since remainder = r, we know: (digit sum − r) must divide exactly by the divisor.
3. Solve for m (or test each answer option systematically).
4. Verify by checking both the digit sum condition AND the last-digits condition (for 4 or 12).

📝 **2024 Q9: THE NUMBER 3MM52 IS DIVIDED BY 9 WITH REMAINDER 2. FIND M.**

Step 1 — Digit sum in terms of m: `3 + m + m + 5 + 2 = 10 + 2m`

Step 2 — Remainder is 2, so (digit sum − 2) must divide by 9:

`(10 + 2m) − 2 = 8 + 2m must divide exactly by 9`

Step 3 — Test: 8 + 2(5) = 18. Does 18 divide by 9? Yes, 18÷9=2. ✓

```
So m = 5. Quick check: 7+5+5+5+2 = 20. 20÷9 = 2 remainder 2 ✓
Answer: m = 5 → (D)
```

📝 **2025 Q9: THE NUMBER 7MM7 IS DIVIDED BY 9 WITH REMAINDER 3. FIND M.**

Digit sum: `7 + m + m + 7 = 14 + 2m`

(14 + 2m) − 3 = 11 + 2m must divide by 9.

```
Test: 11 + 2(8) = 27. 27÷9=3. ✓
Answer: m = 8 → (C)
```

✏️ **PRACTICE QUESTIONS**

1. Is 4 572 divisible by 9?
2. The number 5m34 is divisible by 9 exactly (remainder 0). Find m.
3. The number 2mm6 is divided by 9 with remainder 4. Find m.

(Answers: Yes (4+5+7+2=18), m=7, m=3)

STEP 6 Cryptarithmetic — Letters Instead of Digits

What it is

📖 **WHAT IT IS AND WHY IT'S FUN**

In a cryptarithmetic puzzle, each letter stands for one unique digit from 0 to 9. No two different letters share the same digit. You are shown an addition sum in letter form and must find what digit each letter stands for.

Think of it as a **code-breaking puzzle**. The column-by-column structure of addition is your key. You work from right (ones column) to left, exactly like normal long addition, but in reverse — you are solving for the unknowns that make the addition work.

The key takeaway: This is one column at a time. You never need to guess wildly — the column equation gives you just one or two options, and the carry between columns narrows it further.

The Method

⚡ **THE COLUMN-BY-COLUMN SYSTEM**

1. Start at the **ones column** (rightmost). Write: (sum of all digits in that column) = some number whose ones digit matches the answer's ones digit.
2. Find which digit satisfies this. Note whether there is a **carry** (overflow) into the tens column.
3. Move to the **tens column**. Include the carry. Repeat.
4. Continue left until all letters are found.
5. Write out the complete addition and verify it equals the target.

Mnemonic: "Start right, carry left." Always begin from the rightmost column and carry any overflow to the next.

📝 **2022 Q4: BC + ABC + BC = 692. FIND A × B × C.**

Ones column: C + C + C = 3C. The ones digit of 3C must be 2.

`Test: 3 × 4 = 12. Ones digit is 2 ✓. So C = 4. Carry 1 into tens column.`

Tens column: B + B + B + 1(carry) = 3B + 1. The ones digit must be 9.

`Test: 3 × 6 + 1 = 19. Ones digit is 9 ✓. So B = 6. Carry 1 into hundreds column.`

Hundreds column: A + 1(carry) = 6. So A = 5.

Verify: 64 + 564 + 64 = 692 ✓

`A × B × C = 5 × 6 × 4 = 120 → (D)`

🧠 **THE ONES DIGIT TRICK**

When three identical digits are added (C + C + C = 3C), you only need the last digit to match. List the multiples of 3 and check which ends in your target digit:

- 3×1=3, 3×2=6, 3×3=9, 3×4=**12**, 3×5=15, 3×6=18, 3×7=21, 3×8=24, 3×9=27

Looking for last digit 2: only 3×4=12 works. So C=4. This is very fast once you practise it.

STEP 7 Proportion, Rate and Profit Problems

Type A — Rate and Price per Unit

📖 **WHAT IT IS**

You are given the price for one quantity (e.g. 600g for R108) and asked for the price of a different quantity. The trick is always to find the price for ONE gram (or litre, or unit) first, then multiply by the new amount.

`Price per unit = Total price ÷ Quantity | New price = Price per unit × New quantity`

📝 2024 Q7: IF 600G COSTS R108, WHAT DOES 1 KG COST?

```
Price per gram = 108 ÷ 600 = R0.18
1 kg = 1 000g. New price = 0.18 × 1 000 = R180 → (A)
```

Type B — Profit with Different Buy and Sell Batches

📖 WHAT IT IS AND WHY THE BATCH SIZES MATTER

You buy items in one batch size (e.g. buy 5 for 24 coins) and sell them in a different batch size (sell 6 for 35 coins). Since the batch sizes are different, you cannot compare them directly. You must reduce both to the cost or revenue per single item.

📝 2022 Q10: BUY 5 FOR 24 COINS. SELL 6 FOR 35 COINS. HOW MANY ITEMS TO EARN 620 COINS PROFIT?

```
Cost per item = 24 ÷ 5 = 4.8 coins
Revenue per item = 35 ÷ 6 coins
Profit per item = 35/6 − 24/5 = 175/30 − 144/30 = 31/30 coins
Items needed = 620 ÷ (31/30) = 620 × 30/31 = 20 × 30 = 600 items → (A)
```

⚡ QUICK TRICKS — PROFIT QUESTIONS

- Always find cost-per-item and revenue-per-item before anything else.
- Use a common denominator when subtracting the two fractions.
- "How many items to make a certain profit?" → Target profit ÷ profit per item.

PHASE 3 — GEOMETRY | MONTH 3, WEEKS 9–12

STEP 8 Composite Area (L-shapes and Step Shapes)

What it is

📖 **WHAT IT IS AND WHY IT WORKS**

A composite shape is just two or more rectangles joined together (or one rectangle with a corner cut away). You cannot use a single formula — you must break the shape into simple pieces and add or subtract their areas.

Two approaches — choose the easier one:

- **Add:** Split the L-shape into two rectangles. Calculate each area. Add them.
- **Subtract:** Draw the big surrounding rectangle. Subtract the missing corner piece(s).

The key takeaway: Before calculating anything, find all the missing side lengths. The full outer length minus the inner length gives you the hidden side.

⚡ **THE 4-STEP SYSTEM**

1. Label every given measurement on the figure.
2. Find missing lengths: outer length − inner step = missing piece.
3. Choose Add or Subtract (whichever needs fewer calculations).
4. Calculate each rectangle area and combine.

📝 **2024 Q11: RECTANGLE ABFE HAS 3 TIMES THE AREA OF SQUARE CDGF. HEIGHT OF ABFE = 27M, WIDTH = 36M. PERIMETER OF CDGF?**

```
Area of ABFE = 27 × 36 = 972 m²
Area of square CDGF = 972 ÷ 3 = 324 m²
Side of square = √324 = 18m (check: 18 × 18 = 324 ✓)
Perimeter = 4 × 18 = 72m → (B)
```

STEP 9 Grid Polygon Area — Surround and Subtract

What it is

📖 **WHAT IT IS — AND IT APPEARS EVERY YEAR**

A quadrilateral (4-sided shape) is drawn diagonally across a square grid. Its corners sit on grid points but the sides are diagonal, so you cannot just count squares. The solution is to put a rectangle around it and subtract the corners.

Why it works: The rectangle is easy to calculate. The corners are right-angled triangles. Triangle area = ½ × base × height. Once you subtract all four corners, what remains is the diagonal shape.

⚡ THE 5-STEP SYSTEM

1. Read off the coordinates of all four corners from the grid carefully. Count grid squares.
2. Draw (or imagine) the smallest rectangle that fully encloses all four corners.
3. Calculate: rectangle width × rectangle height.
4. For each corner of the rectangle: find the right triangle that sits outside the quadrilateral. Calculate ½ × base × height.
5. Answer = rectangle area − sum of all four triangle areas.

Mnemonic: "Box it in, cut the corners."

📝 GENERAL EXAMPLE: BOUNDING RECTANGLE 5×4 = 20 CM²

```
Corner triangles: ½×2×1=1, ½×3×2=3, ½×2×3=3, ½×1×2=1
Total corners = 1+3+3+1 = 8 cm²
Quadrilateral area = 20 − 8 = 12 cm²
```

The 2022 Q13 answer is (C) 12 and 2023 Q13 is (C) 14. Practise by drawing the grids from each paper.

STEP 10 Lines of Symmetry

What it is

📖 WHAT IT IS

A shape has a line of symmetry if you can fold it in half exactly and both halves are perfect mirror images. Lines can be horizontal, vertical, or diagonal (at 45°).

These questions give you a base shape (made of joined squares) and ask you to place one extra square to **maximise** the number of symmetry lines in the combined shape.

⚡ QUICK METHOD

1. Draw the base shape on squared paper.
2. Try attaching the extra square to each possible edge — one trial at a time.
3. For each attempt, fold the paper (or use a ruler as a mirror) along every possible axis: horizontal, vertical, and diagonal.
4. The position that folds perfectly in the most ways is your answer.

Tip: Shapes with the most symmetry look "balanced" in every direction. Usually, placing the extra square at the centre of an edge (rather than at a corner) adds the most symmetry.

STEP 11 Cube Tower Patterns — Triangular Numbers in 3D

What it is

📖 **WHAT IT IS AND WHY IT USES THE GAUSS FORMULA**

The tower shown in the papers is a 3D staircase of small cubes. Looking from the top, each storey is a staircase arrangement. Storey 1 (top) = 1 cube. Storey 2 = 3 cubes (1+2). Storey 3 = 6 cubes (1+2+3). Storey k = 1+2+3+...+k = k(k+1)/2 cubes — the triangular number formula you learned in Foundation Concept B!

To find the total for an n-storey tower, add up all the triangular numbers from storey 1 to storey n. A table is the fastest way to do this in the exam.

📝 **BUILD THE TABLE — ALWAYS WORKS IN THE EXAM**

Storey (from top)	Formula k(k+1)÷2	Cubes	Running total
1	1×2÷2	1	1
2	2×3÷2	3	4
3	3×4÷2	6	10
4	4×5÷2	10	20
5	5×6÷2	15	35
6	6×7÷2	21	56
7	7×8÷2	28	84
8	8×9÷2	36	120
9	9×10÷2	45	165
10	10×11÷2	55	220

2024 Q12: 10-storey tower = **220 cubes** → (C)

2023 Q10: Last storey has 300 cubes. Find storeys: k(k+1)/2 = 300 → k(k+1) = 600 → k=24 (24×25=600 ✓) → **24 storeys** → (A)

2025 Q12: 4-storey total = 20. Nearest cube bigger than 20: $4^3 = 64$. Add 64−20 = **44 cubes** → (A)

🧠 **BUILD THE TABLE, DON'T MEMORISE THE FORMULA**

You do NOT need to memorise the formula for the total of an n-storey tower. Just build the table above, row by row. It takes about 90 seconds and is completely reliable. Practice building it until you can do it in under 2 minutes.

STEP 12 Age Problems — Three Types

What it is

📖 **WHAT IT IS AND WHY ALGEBRA IS THE KEY**

Age problems always tell you how two people's ages are related, plus a total or future age. The relationship is most clearly expressed as an algebra equation. Choose one person's age as your unknown (call it n), write the other person's age in terms of n, then use the combined condition to form an equation and solve.

The key takeaway: always give the younger person's age the letter n. Then express the older person's age in terms of n. The combined sum gives you the equation.

Type A — Direct Relationship (One person described in terms of the other)

📝 **2024 Q14: ROMY IS 3 YEARS MORE THAN HALF THE MOTHER'S AGE. MOTHER WILL BE 50 IN 8 YEARS.**

```
Mother's current age = 50 − 8 = 42
Romy = (1/2 × 42) + 3 = 21 + 3 = 24 → (C)
```

📝 **2025 Q14: TINA IS 3 YEARS LESS THAN HALF THE FATHER'S AGE. FATHER WILL BE 56 IN 10 YEARS.**

```
Father's current age = 56 − 10 = 46
Tina = (1/2 × 46) − 3 = 23 − 3 = 20 → (C)
```

Type B — "X is N times what Y was M years ago" + combined sum (Learn this one well!)

📝 **2023 Q12: BUNNY IS TWICE THE AGE AMY WAS 10 YEARS AGO. COMBINED CURRENT AGES = 55. FIND AMY'S AGE.**

Let Amy's current age = A (our unknown).

```
Amy was (A − 10) years old ten years ago.
Bunny's current age = 2 × (A − 10) = 2A − 20
```

Combined sum condition:

```
A + (2A − 20) = 55
3A − 20 = 55
3A = 75
A = 25
Check: Bunny = 2(25−10) = 30. Sum = 25+30 = 55 ✓ → (B)
```

⚡ **THE AGE PROBLEM CHECKLIST**

- Find any current age given a future age: current = future − years to go.
- Let the younger person's age = A.
- Write the older person's age as a formula involving A.
- Use the combined sum to write one equation. Solve for A.
- Always check both the relationship AND the sum are satisfied.

Note on 2022 Q12: The algebra gives a non-integer result — this appears to be an error in the original paper. The official answer is (C) 25. Always check www.saolympiads.co.za for the official answer key.

STEP 13 Hotel Room Number-Line Positioning

What it is

📖 **WHAT IT IS AND THE ONE THING MOST STUDENTS MISS**

Hotel rooms are numbered in a consecutive sequence — all odd (1, 3, 5, 7...) or all even (2, 4, 6, 8...). A worker moves along this sequence by a certain number of rooms in each direction. You must find the starting room.

The one thing most students miss: moving one room right in an odd-numbered hotel means the room number increases by 2 (not 1), because consecutive odd numbers differ by 2. So "4 rooms right" = room number +8.

🧠 **PICTURE THE HOTEL**

Imagine walking down a hotel corridor. Room 1, Room 3, Room 5, Room 7... Every door you pass, the number goes up by 2. So if you walk past 5 doors, the room number has gone up by 5 × 2 = 10.

Formula: 1 room move = 2 room numbers.

📝 **2023 Q15: ODD ROOMS. MOVES: +5 RIGHT, −9 LEFT, +1 RIGHT. ENDS AT ROOM 5. FIND START.**

Let S = starting room number.

Each move of r rooms = change of 2r in room number.

```
S + 2(5) − 2(9) + 2(1) = 5
S + 10 − 18 + 2 = 5
S − 6 = 5
S = 11
Check: 11 is odd ✓. Starting at 11: +10→21, −18→3, +2→5 ✓ → (B)
```

⚡ **QUICK TRICKS — HOTEL ROOMS**

- **Formula: S + 2×(net moves) = final room.** Net moves = sum of all moves (right = positive, left = negative).
- Your final answer must be odd (if odd hotel) or even (if even hotel). If not — check your signs.
- Alternatively: work backwards. Undo the last move first, then the second-to-last, and so on.
- **Mnemonic: "One room, two numbers."**

STEP 14 Strike-Off Sequences

What it is

📖 **WHAT IT IS**

A list of consecutive numbers is written out. In round 1, every nth number is crossed off. In round 2, from what remains, every mth number is crossed off. You must count the survivors with a specific property (odd, even, or prime).

The secret is to work with a **small representative group** (of size LCM(n, m)), find the survivor pattern for that group, then scale up to cover all numbers.

⚡ **THE GROUP METHOD — STEP BY STEP**

1. Find LCM(n, m). This is the size of one repeating group.
2. List positions 1 to LCM in your group. Apply round 1 strike-off.
3. Apply round 2 strike-off to the survivors of round 1.
4. Count survivors in this one group = s.
5. Total survivors = (total numbers ÷ LCM) × s.
6. From survivors, count only those with the required property.

Mnemonic: "LCM is the group size. Find the pattern once. Scale it up."

📝 **2024 Q18: NUMBERS 11–80 (70 NUMBERS). STRIKE EVERY 3RD, THEN EVERY 4TH. COUNT ODD SURVIVORS.**

```
LCM(3,4) = 12. Work with one group of 12 positions.
Round 1 (strike every 3rd): cross positions 3, 6, 9, 12.
Survivors: 1, 2, 4, 5, 7, 8, 10, 11 (8 survivors).
Round 2 (strike every 4th survivor): cross 4th (pos 5) and 8th (pos 11).
Final survivors: 1, 2, 4, 7, 8, 10 (6 per group of 12).
70 numbers = 5 full groups + 10 remainder.
From full groups: 5 × 6 = 30 survivors.
```

(Map positions to actual numbers starting at 11, count odd ones to get the final answer.)

`Official answer: ≈14 odd numbers`

STEP 15 Watch Loses or Gains Time

What it is

📖 **WHAT IT IS AND WHY THERE ARE TWO TIME-STREAMS**

A faulty watch that loses X minutes per hour runs too slowly. Picture two clocks side by side: real time ticks normally (60 minutes per hour), but the watch only ticks (60 – X) minutes in the same real hour. They are proportional — two different speeds, same starting point.

The key takeaway: real time and watch time form a ratio. If the watch has advanced W minutes, you can find how many real minutes R have passed using the ratio.

```
Real minutes = Watch minutes × 60 ÷ (60 − X)
```

```
Watch minutes = Real minutes × (60 − X) ÷ 60
```

🧠 **THE TWO-CLOCK PICTURE**

Imagine two clocks on a wall. The left clock (real time) ticks 60 times per hour. The right clock (the broken watch) only ticks 55 times per hour if it loses 5 min/hour. After 660 right-clock ticks, the real clock has ticked 660 × (60/55) = 720 times. That's 12 real hours.

Mnemonic: "Multiply up by 60, divide by what the watch actually shows per hour."

📝 **2022 Q17: WATCH LOSES 10 MIN/HOUR. CORRECT AT 08:00. WATCH SHOWS 13:00. ACTUAL TIME?**

```
Watch advanced = 13:00 − 08:00 = 5 hours = 300 watch-minutes.
Real minutes = 300 × 60 ÷ (60−10) = 300 × 60/50 = 300 × 6/5 = 360 real minutes = 6 hours.
Actual time = 08:00 + 6h = 14:00
```

📝 **2025 Q13: WATCH LOSES 5 MIN/HOUR. CORRECT AT 06:00. WATCH SHOWS 17:00. ACTUAL TIME?**

```
Watch advanced = 11 hours = 660 watch-minutes.
Real minutes = 660 × 60/55 = 660 × 12/11 = 720 real minutes = 12 hours.
Actual time = 06:00 + 12h = 18:00 → (B)
```

STEP 16 Plant Geometric Growth — Multiplying Sequences

What it is

📖 **WHAT IT IS AND HOW IT DIFFERS FROM ARITHMETIC SEQUENCES**

An arithmetic sequence adds the same amount each step. A **geometric sequence** multiplies by the same amount each step. That multiplier is called the **common ratio**.

For example: 8, 16, 32, 64 ... Each term is the previous one multiplied by 2. The ratio is 2.

Language trap: "Grows by 1/3 of its height" does NOT mean multiply by 1/3. It means: new height = old + (1/3 × old) = old × (1 + 1/3) = old × 4/3. The ratio is 4/3.

To go BACKWARDS one step: divide by the ratio. Dividing by 4/3 = multiplying by 3/4 (flip the fraction).

🧠 **THE "GROWS BY" LANGUAGE TRICK**

- "Grows by half its height" → new = old × (1 + 1/2) = old × **3/2**
- "Grows by one-third of its height" → new = old × (1 + 1/3) = old × **4/3**
- "Grows by one-quarter of its height" → new = old × **5/4**

Mnemonic: "Add 1 to the fraction to get the ratio." 1 + 1/2 = 3/2. 1 + 1/3 = 4/3.

📝 **2025 Q19: PLANT GROWS BY ADDING 1/3 OF ITS HEIGHT EACH WEEK. WEEK 4 = 64CM. FIND WEEK 1.**

```
Growth ratio = 1 + 1/3 = 4/3. Reciprocal (to go backward) = 3/4.
Week 4 = 64.
Week 3 = 64 × 3/4 = 48.
Week 2 = 48 × 3/4 = 36.
Week 1 = 36 × 3/4 = 27 cm
```

⚡ **QUICK TRICKS — PLANT GROWTH**

- Identify the ratio from the "grows by" language. Add 1 to the fraction.
- Going backwards: multiply by the reciprocal (flip the fraction) for each step back.
- Check: going forwards from Week 1 should give back the Week 4 value.

STEP 17 Sum of Cubes and Telescoping Series

Type A — Sum of Cubes

📖 WHAT IT IS AND WHY IT'S A BEAUTIFUL SHORTCUT

You need to add a long list of cube numbers: $1^3 + 2^3 + 3^3 + \ldots + n^3$. Calculating each cube and adding them one by one would take forever. The shortcut is a stunning mathematical fact:

The sum of the first n cubes equals the SQUARE of the Gauss sum.

You already know the Gauss sum (Foundation Concept B): 1+2+...+n = n(n+1)/2. So the sum of cubes is just that value, squared.

$$1^3 + 2^3 + 3^3 + \ldots + n^3 = [\, n(n+1) \div 2 \,]^2$$

🧠 VERIFY IT BEFORE TRUSTING IT

```
n=2: 1³+2³ = 1+8 = 9. [2×3÷2]² = 3² = 9. ✓
n=3: 1³+2³+3³ = 36. [3×4÷2]² = 6² = 36. ✓
n=4: 1+8+27+64 = 100. [4×5÷2]² = 10² = 100. ✓
```

Every time the formula works perfectly. Once you have verified it for small n, you can use it confidently for large n.

📝 2024 Q20: CALCULATE $1^3 + 2^3 + \ldots + 20^3$

```
Gauss sum for n=20: 20 × 21 ÷ 2 = 210.
Sum of cubes = 210² = 44 100
```

📝 2023 Q20: CALCULATE $3^3 + 4^3 + \ldots + 24^3$

Step 1 — Full sum $1^3+\ldots+24^3$: `[24×25÷2]² = 300² = 90 000`

Step 2 — Remove 1^3+2^3: `1 + 8 = 9`

```
Answer = 90 000 − 9 = 89 991
```

Type B — Telescoping Fractions

📖 WHAT IT IS — AND WHY THE CANCELLATION HAPPENS

A telescoping series is a long sum of fractions where most terms cancel out, leaving only the very first piece and the very last piece. It is like a collapsing telescope — many sections fold away and you are left with just two ends.

The key is using a **partial fraction identity** to split each term into two simpler fractions. After splitting, you will see that consecutive terms share the same sub-fractions with opposite signs — and they cancel.

```
1/(n × (n+2)) = (1/2) × (1/n − 1/(n+2))
```

🧠 **PROVE THE IDENTITY YOURSELF — TAKES 30 SECONDS**

Take the right side: (1/2)(1/n – 1/(n+2)). Find a common denominator of n(n+2):

```
= (1/2) × [(n+2 − n) / n(n+2)]
= (1/2) × [2 / n(n+2)]
= 1 / n(n+2) ✓
```

It matches the left side exactly. The identity is proved. Now use it with confidence.

📝 **2022 Q20: 1/(3×5) + 1/(5×7) + 1/(7×9) + … + 1/(19×21)**

Split every term using the identity:

```
1/(3×5) = (1/2)(1/3 − 1/5)
1/(5×7) = (1/2)(1/5 − 1/7)
1/(7×9) = (1/2)(1/7 − 1/9)
…
1/(19×21) = (1/2)(1/19 − 1/21)
```

Now add. Notice what cancels: −1/5 and +1/5 cancel. −1/7 and +1/7 cancel. And so on.

```
Only the very first (+1/3) and very last (−1/21) survive:
Sum = (1/2)(1/3 − 1/21) = (1/2)(7/21 − 1/21) = (1/2)(6/21) = 1/7
```

STEP 18 Worst-Case Probability — Blindfolded Disc Problems

What it is

📖 **WHAT "ENSURE" MEANS — THE MOST IMPORTANT WORD IN THE QUESTION**

The word **ENSURE** means you must guarantee the result no matter how badly things go. You cannot assume luck. You must assume the **worst possible luck** with every single pick.

Worst case means: every disc you pick, until the very end, is from the wrong colour. Only at the very last picks do you start getting the colour you want.

The key takeaway: to guarantee K discs of the target colour, first assume you pick ALL the non-target discs, then pick K target discs.

```
Minimum picks to ENSURE K of target colour = All non-target discs + K
```

🧠 **WHY NOT JUST K?**

Imagine you want 4 white discs from a bag with 6 white, 4 green and 3 orange. Could you just pick 4? In theory — but you might pick green, green, orange, green (4 picks, zero white). You might pick 7 non-white discs before your first white one. Once all 7 non-white are gone, the next 4 must be white. So minimum = 7 + 4 = 11.

📝 **2023 Q8: 6 WHITE, 4 GREEN, 3 ORANGE. ENSURE 4 WHITE DISCS.**

```
Non-target discs = 4 green + 3 orange = 7.
Minimum picks = 7 + 4 = 11 → (D)
```

📝 **2024 Q10: 6 WHITE, 7 GREEN, 4 ORANGE. ENSURE 3 GREEN DISCS.**

```
Non-target discs = 6 white + 4 orange = 10.
Minimum picks = 10 + 3 = 13 → (C)
```

STEP 19 Three-Variable Equations — Basketball Scoring

What it is

📖 **WHAT IT IS AND THE SIMPLE SUBSTITUTION STRATEGY**

When a problem has three unknown quantities, you need three pieces of information (three equations). The strategy is **substitution**: find the simplest equation (usually one that says "a = 2b" or "b = 2a"), substitute it everywhere, and reduce from 3 unknowns → 2 unknowns → 1 unknown → solve.

⚡ **THE 7-STEP SYSTEM**

1. Name the three unknowns: a, b, c. Write what each represents.
2. Write three equations from the three given pieces of information.
3. Pick the simplest equation (often the ratio/relationship one).
4. Substitute that into the other two equations. Now you have two equations with two unknowns.
5. From one of those, express one variable in terms of the other. Substitute into the last equation.
6. Solve the one-variable equation. Work backwards for the others.
7. Check ALL THREE values in ALL THREE original equations.

📝 2022 Q18: 95 POINTS FROM 45 SHOTS IN 1S, 2S AND 3S. TWICE AS MANY 2-POINTERS AS 1-POINTERS. FIND 3-POINTERS.

```
Let a = 1-pointers, b = 2-pointers, c = 3-pointers.
Eq1 (shots): a + b + c = 45
Eq2 (points): a + 2b + 3c = 95
Eq3 (ratio): b = 2a
```

Sub b=2a into Eq1: `a + 2a + c = 45 → 3a + c = 45 ... (i)`

Sub b=2a into Eq2: `a + 4a + 3c = 95 → 5a + 3c = 95 ... (ii)`

From (i): c = 45 – 3a. Sub into (ii):

```
5a + 3(45-3a) = 95 → 5a + 135 - 9a = 95 → -4a = -40 → a = 10
c = 45 - 30 = 15. b = 20.
```

Check: 10+20+15=45 ✓ 10+40+45=95 ✓ 20=2×10 ✓

```
3-pointers = 15
```

STEP 20 Digit Patterns and Last-Digit Problems

Type A — Last N digits of a large product

📖 WHAT IT IS AND WHY YOU CAN IGNORE THE BIG DIGITS

When you multiply two large numbers, the last 3 digits of the answer depend only on the last 3 digits of each number. The hundreds of thousands and millions contribute zero to the ones, tens, or hundreds columns. So you can discard everything above the last 3 digits and work with much smaller numbers.

📝 2022 Q11: LAST 3 DIGITS OF 77 777 × 99 999

Use the factoring trick: 77 777 × 99 999 = 77 777 × (100 000 – 1)

```
= 7 777 700 000 - 77 777 = 7 777 622 223
Last 3 digits: 223 → (C)
```

Type B — Digit sum patterns

📝 2023 Q18: FIND DIGIT SUM OF 55 555 555 × 99 999 999

```
55 555 555 × 99 999 999 = 55 555 555 × (100 000 000 - 1)
= 5 555 555 500 000 000 - 55 555 555
= 5 555 555 444 444 445
Digits: eight 5s and eight 4s.
Digit sum = 8×5 + 8×4 = 40 + 32 = 72
```

PHASE 6 — PAPER PRACTICE | MONTH 6, WEEKS 21–26

Weeks 21–22: Timed Papers 2022 and 2023

Sit each paper under full exam conditions — 90 minutes, no calculator, no notes, no help. Mark immediately after. Sort your mistakes into two categories: **careless errors** (you knew the method but slipped on arithmetic) and **concept gaps** (you didn't know the method). Careless errors are fixed by slowing down and checking. Concept gaps are fixed by revisiting the relevant step in this booklet.

Weeks 23–24: Timed Papers 2024 and 2025

Same process. Pay extra attention to topics that appear in both papers: cube towers, plant growth, related rectangles, hotel room positioning, watch time.

Week 25: Mixed Topic Sprint

From all four marked papers, write a list of your three weakest topics. Spend one full session on each of those three topics, drawing questions from all four years.

Week 26: Final Revision and Confidence Check

Redo every question you got wrong in practice. Go through the Quick Reference Card. Try to complete Q1–Q10 of any paper in under 30 minutes. You are ready.

EXAM DAY MINDSET

You have worked for 6 months. You have seen every question type before. When you sit down in the exam:

- Read Q1 twice before writing. Get it right. It is a free mark.
- If you are stuck, skip and come back. Do not stall on one question.
- Show all working. Even if your answer is wrong, working earns marks.
- Trust the method. If you have used the right method, the answer is correct.

BONUS — 2017 FINAL ROUND EXTENSION TOPICS

Harder than the standard paper. Only attempt these after all 20 steps above are fully secure.

Bonus A: Handshakes

📝 2017 Q14: EACH PERSON SHOOK HANDS WITH 1/4 OF THE PEOPLE. TOTAL = 288 HANDSHAKES. HOW MANY PEOPLE?

Each handshake involves 2 people: $n \times (n/4) \div 2 = 288 \rightarrow n^2/8 = 288 \rightarrow n^2 = 2304 \rightarrow n = 48$ ✓

Answer: 48 people

Bonus B: Two-Tap Filling

📝 2017 Q19: COLD TAP FILLS 140L IN 15 MIN. HOT TAP IN 20 MIN. BOTH OPEN — HOW LONG?

Rates: $140/15 + 140/20 = 140 \times (1/15 + 1/20) = 140 \times 7/60$ L/min.

Time $= 140 \div (140 \times 7/60) = 60/7 =$ 8 and 4/7 minutes

Bonus C: Painted Cube Faces

📝 2017 Q20: 125 SMALL CUBES → 5×5×5 LARGE CUBE, ALL FACES PAINTED. EXACTLY 2 GREEN FACES?

Edge cubes (not at corners): 12 edges × 3 interior cubes per edge = 36 cubes

Check: 8 corner (3 faces) + 36 edge (2 faces) + 54 face (1 face) + 27 interior (0) = 125 ✓

Bonus D: Fibonacci-Style Sequences

📝 2017 Q1: 3; 4; 7; 11; 18; … (EACH TERM = SUM OF PREVIOUS TWO). FIND THE 8TH TERM.

T1=3, T2=4, T3=7, T4=11, T5=18, T6=29, T7=47, T8=29+47=76

Bonus E: Lines Dividing a Plane

📝 2025 Q20: 2 LINES → 4 REGIONS. 3 LINES → 7 REGIONS. HOW MANY LINES FOR 92 REGIONS?

Each new line adds one more region than the previous line: new regions added are 2, 3, 4, 5, …

Total regions with n lines $= 1 + (2+3+4+\ldots+(n+1)) = n(n+1)/2 + 1$.

Check: n=2: 4 ✓. n=3: 7 ✓.

Solve: $n(n+1)/2 + 1 = 92 \rightarrow n(n+1) = 182 \rightarrow n=13$ ($13 \times 14 = 182$ ✓)

Answer: 13 lines

Online Resources — When and How to Use Them

Code	Resource	Address	Best for
KA	Khan Academy	khanacademy.org	Video lessons + graded practice on all topics
CM	Corbett Maths	corbettmaths.com	Short videos + printable worksheets with answers
MF	Math is Fun	mathsisfun.com	Clear explanations + interactive examples
AoPS	Art of Problem Solving	artofproblemsolving.com/wiki	Olympiad-level theory
NR	NRICH	nrich.maths.org	Olympiad-style challenge problems
EW	Eddie Woo	youtube.com/@misterwootube	Engaging video explanations
SA	SA Olympiads (official)	saolympiads.co.za	Past papers + official answer sheets

Month 1 (Weeks 1–4) — Foundation Repair

Week 1: KA — "Order of Operations" — 3 videos + all practice. CM — "BIDMAS" — worksheet + self-mark.

Week 2: CM — "Decimal Division." MF — "Long Division" page — read + 5 problems.

Week 3: KA — "Fractions of a whole" + "Solving equations with fractions." MA — "Mathantics Algebra Basics" on YouTube.

Week 4: EW — "Eddie Woo factoring expressions." AoPS Wiki — "Algebraic manipulations."

Month 2 (Weeks 5–8) — Core Number

Week 5: KA — "Divisibility tests" — all 5 practice levels. MF — "Divisibility Rules" — memorise + quiz.

Week 6: CM — "Divisibility" worksheet. NR — "Divisible digits."

Week 7: AoPS — "Alphametics" — read + 2 examples. NR — "Digit puzzles."

Week 8: KA — "Unit rates" + "Solving proportions." CM — "Ratio and proportion problems."

Month 3 (Weeks 9–12) — Geometry

Week 9: KA — "Area of composite figures." CM — "Area of compound shapes."

Week 10: MF — "Pick's Theorem." EW — "Area of irregular shapes on grids."

Week 11: KA — "Lines of symmetry." NR — "Triangular numbers."

Week 12: EW — "Sum of first n integers" (Gauss trick). SA — attempt 2023 Q10–Q13.

Month 4 (Weeks 13–16) — Logic

Week 13: KA — "Age word problems." NR — "Age problems."

Week 14: Draw odd-numbered rooms 1,3,5,...,21 on a number line. Simulate 2023 Q15 using Step 13 method. CM — "Solving linear equations."

Week 15: NR — "Sieve of Eratosthenes." Build your own strike-off table for 1–30.

Week 16: KA — "Geometric sequences." EW — "Geometric sequences."

Month 5 (Weeks 17–20) — Advanced

Week 17: EW — "Sum of natural numbers" (Gauss). MF — "Triangular numbers."

Week 18: MF — "Sum of cubes." AoPS — "Telescoping series."

Week 19: KA — "Systems of 3 equations." NR — combinatorics problems.

Week 20: NR — Stage 3 challenges. SA — full 2017 Final Round (120 minutes, timed).

Month 6 (Weeks 21–26) — Paper Practice

After each timed paper: look up ONLY the topics you got wrong. Revisit that step in this booklet. Do NOT learn new topics. Reinforce what you already know. SA saolympiads.co.za is your primary source for past papers and answer sheets.

Quick Reference Card — Print and Keep

Topic	Formula or Key Rule	Quick memory hook
BODMAS	Brackets → Orders → ÷× (left to right) → +− (left to right)	*"Big Orange Dogs Might Act Strange"*
Rounding	Look one place past target: ≥5 rounds up, ≤4 stays	*"5 and above, round it up"*
Divide by decimal	Multiply both by 10 or 100: 8÷0.04 = 800÷4	*"Kick the decimal away"*
Container fractions	Capacity = Amount ÷ (big fraction − small fraction)	*"Amount over gap"*
Fraction midpoint	x = 2 × Middle − Other end	*"Double the middle, minus the end"*
Distributive law	a×b + a×c = a×(b+c) — factor out the common number	*"Same number? Factor it out!"*
Gauss sum 1 to n	n(n+1) ÷ 2	*"Pair the ends, count the pairs, halve"*
Arithmetic series	(terms) × (first + last) ÷ 2	*"Average the ends, times the count"*
Sum of cubes	$1^3+2^3+...+n^3 = [n(n+1)÷2]^2$	*"Gauss sum, then square it"*
Telescoping identity	1/(n×(n+2)) = (1/2)(1/n − 1/(n+2))	*Split, cancel the middles, keep the ends*
Divisibility by 9	Sum of all digits divisible by 9	*"Digit Sum Club: 3 and 9 are members"*
Age Type B	Let younger = A. Older = k(A−m). Sum equation. Solve.	*"Younger first, then express the older"*
Hotel rooms	S + 2×(each move signed) = final room	*"One room, two numbers"*
Watch loses X min/hr	Real time = Watch time × 60 ÷ (60−X)	*"Multiply up by 60"*
Plant grows by 1/k	Ratio = 1 + 1/k. Go back: multiply by reciprocal.	*"Add 1 to the fraction to get the ratio"*
Worst-case discs	All non-target + required target	*"Worst luck first, then your colour"*
Cube tower storey k	k(k+1)÷2 cubes. Build the table.	*"Triangular number for each storey"*
Lines dividing a plane	n(n+1)/2 + 1 regions	*"Gauss sum plus 1"*

🧠 **THE EQUATION CHECKLIST — USE EVERY TIME**

1. Name the unknown. Write "Let n = ..."
2. Translate every sentence into algebra.
3. Write the equation with a clear equals sign.
4. Solve one step per line. Show all steps.
5. Check: substitute your answer back into the original problem.

⏱ TIME BUDGET FOR THE EXAM

Questions	Time each	Total
Q1–Q10 (multiple choice)	2–3 minutes	25 minutes
Q11–Q15 (multiple choice)	4–5 minutes	20 minutes
Q16–Q20 (open-ended, 2 marks)	8–9 minutes	45 minutes
Total		**90 minutes**

Finish early? Go back and check Q1–Q10 first. One-mark careless errors are the most common way to lose a ranking position.

Source: FEMSSISA Grade 6 Olympiad papers 2022, 2023, 2024, 2025 and 2017 Final Round. All worked examples verified against official papers. Certainty: 9/10. The 2022 Q12 age problem produces a non-integer algebraic result and appears to contain an error in the original paper — consult the official answer key at **www.saolympiads.co.za**. This document was designed for clean black-on-white printing. Open in any browser and press Ctrl+P.